

Psych Medication Checkpoints - Antipsychotics and Mood Stabilizers

Category	Medication	Clinical Pearls	Black Box Warnings	Psych C&L Management
First Generation Antipsychotics	Chlorpromazine (Thorazine ®)	Lower potency - Sedation, weight gain, anticholinergic activity Comes as tablet and short acting IM injection Use lower IM doses because oral first-pass metabolism is significant (less is absorbed when taken orally)	Increased mortality in elderly patients with dementia-related psychosis	- See LAI chart for additional information about dosing Labs/Diagnostics: <u>QTc - significant QTc prolongation***</u> - Orthostatic blood pressure checks
	Fluphenazine (Prolixin ®)	High potency - EPS Comes as tablet, oral liquid, short acting injection (HCl) and long acting injection (decanoate)		- See LAI chart for additional information about dosing Labs/Diagnostics: - QTc - potential for QTc prolongation - Orthostatic blood pressure checks
	Haloperidol	High potency - EPS CYP 2D6 Comes as tablet, oral liquid, short acting injection, and long acting injection (decanoate)		Labs/Diagnostics: <u>QTc - significant QTc prolongation with IV***</u> - Orthostatic blood pressure checks
	Loxapine (Loxitane ®)	High potency - EPS Two active metabolites - complex pharmacokinetics Comes as tablet and aerosol powder Aerosol formulation - reaches peak plasma concentration in 2 minutes		Labs/Diagnostics: - QTc - potential for QTc prolongation - Orthostatic blood pressure checks REMS for the potential of bronchospasms for inhaled loxapine
	Perphenazine (Trilafon ®)	Moderate potency - EPS Works in 1-3 hours CYP 2D6 Comes as tablets only		Labs/Diagnostics: - QTc - potential for QTc prolongation - Orthostatic blood pressure checks
	Pimozide (Orap ®)	High potency - EPS Does not have indication for schizophrenia		Labs/Diagnostics: <u>QTc - significant QTc prolongation***</u>

		Can be used off label or to treat delusional disorders Avoid CYP1A2 or 3A4 inducers or inhibitors Get CYP2D6 genotype if doses greater than 4 mg are used daily Comes as tablets only		Orthostatic blood pressure checks
	Thioridazine (Mellaril ®)	Lower potency - Sedation, weight gain, anticholinergic activity Pigmentary retinopathy Comes as tablets only		Labs/Diagnostics: <u>QTc - avoid use if QTc is over 450***</u> - Orthostatic blood pressure checks
	Thiothixene (Navane ®)	High potency - EPS Comes as capsules only		Smoking status - consider decreasing home dose inpatient
	Trifluoperazine (Stelazine ®)	Intermediate potency - EPS Comes as tablets only		Labs/Diagnostics: - QTc - potential for QTc prolongation - Orthostatic blood pressure checks
Second Generation Antipsychotics	Aripiprazole (Abilify ®, Aristada ®, Abilify Maintena ®)	Highest rate of akathisia Impulsive behaviors - assess in patients with a history of gambling, substance use, etc. Comes as long acting injectables (Aristada and Abilify Maintena), oral solution (1 mg/mL), tablets and ODTs	Increased mortality in elderly patients with dementia-related psychosis Increased risk of suicidal thoughts and behavior in pediatric and young adult patients	- Monitor for akathisia - Impulsive behaviors - assess in patients with a history of gambling, substance use, etc. - See LAI chart for additional information about dosing Abilify Maintena ® and Aristada ® / Aristada Initio ® Labs: - A1c - BMI
	Asenapine (Saphris ® and Secuado ®)	Dissolves under the tongue - do not eat, drink, or smoke within 10 minutes of taking the medication Medication must be dissolved and not swallowed - poor GI absorption Comes as patches and SL tablets - brand		Labs/Diagnostics: - A1c - BMI Other checkpoints: - Smoking status

		name only		
	Brexpiprazole (Rexulti ®)	Akathisia Only comes as tablets, only brand name (Rexulti ®)	Increased mortality in elderly patients with dementia-related psychosis	• Monitor for akathisia Labs: • A1c • BMI
	Cariprazine (Vraylar ®)	LONG half life (7-21 <u>days</u> for active metabolites) - may take weeks to see dose change effects Only comes as capsules - brand name only	Increased risk of suicidal thoughts and behavior in pediatric and young adult patients	Labs: • A1c • BMI
	Clozapine (Clozaril ®, FazaClo ®, Versacloz ®)	Anticholinergic (recommend scheduled bowel regimen) Smoking status to determine if adjustments need to be made inpatient Monitor for metabolic effects Comes as tablets, ODT, and oral solution	Agranulocytosis Seizures Myocarditis and Cardiomyopathy Orthostatic Hypotension Increased Mortality in Elderly Patients	Labs/Diagnostics (UTILIZE CLOZAPINE ORDER SET) • BMI • A1c • Seizure precautions • Orthostatic blood pressure checks • Absolute neutrophil count (ANC) ○ Clozapine REMS - enter levels once weekly to start • Can draw a level to determine compliance - this is a send-out lab that takes awhile to result ○ Target level - 200-700 Other checkpoints: • Smoking status - may need to adjust dose based on ADRs • Scheduled bowel regimen
	Iloperidone (Fanapt ®)	Dose related/limiting orthostatic hypotension Only comes as tablets	Increased mortality in elderly patients with dementia-related psychosis	Labs: • A1c • BMI
	Lumateperone (Caplyta ®)	Only dosing is 42 mg once daily No renal or hepatic adjustments - use with caution Only comes as capsule - brand name only	Increased mortality in elderly patients with dementia-related psychosis	Give with food Labs: • A1c • BMI

	Lurasidone (Latuda ®)	Only comes as tablets - brand name only	Increased mortality in elderly patients with dementia-related psychosis Increased risk of suicidal thoughts and behavior in pediatric and young adult patients	- When putting the order in, make sure to specify it needs to be taken with a substantial snack or meal (350 calories) Labs: - A1c - BMI - SCr Dose-related serum creatinine increase
	Olanzapine (Zyprexa ®, Zydus ®, Zyprexa Relprevv ®)	<u>Metabolic side effects</u> Comes as tablet, ODT, short acting IM, long acting IM (Relprevv ®)	Increased mortality in elderly patients with dementia-related psychosis Post Injection delirium/sedation syndrome (for Zyprexa Relprevv)	- See LAI chart for additional information about dosing - Smoking status - may need to decrease home dose - Smokers may require 30% greater daily dose - Concomitant use of parenteral benzodiazepines - Respiratory depression - hard stop at UH Labs: - A1c - BMI
	Paliperidone (Invega ®, Invega Sustenna ®, Invega Trinza ®)	Metabolic side effects Comes as ER tablets, one month LAI, 3 month LAI	Increased mortality in elderly patients with dementia-related psychosis	See LAI chart for additional information about dosing Ghost tablet in stool - remind patients/nurses Labs: - May increase prolactin - monitor for symptoms of hyperprolactinemia - A1c - BMI
	Quetiapine (Seroquel ®)	<u>Metabolic side effects</u> ER tablets NOT to be given with food - only food less than 300 calories Comes as tablets and ER tablets	Increased mortality in elderly patients with dementia-related psychosis Increased risk of suicidal thoughts and behavior in pediatric and young adult	Labs/Diagnostics: - QTc - potential for QTc prolongation - A1c - BMI

			patients	
	Risperidone (Risperdal ®, Risperdal M-tab ®, Risperdal Consta ®, Perseris ®)	Metabolic side effects Comes as tablets, ODT, oral solution, IM (Consta ®), subcutaneous (Perseris ®)	Increased mortality in elderly patients with dementia-related psychosis	See LAI chart for additional information about dosing Labs: ● Prolactin ● A1c ● BMI Lower initial dose for patients with CrCl under 30 mL or severe hepatic impairment Floppy iris syndrome
	Ziprasidone (Geodon ®)	QTc prolongation Comes as capsules and IM injection		Labs/Diagnostics: ● <u>QTc***</u> Other checkpoints ● When putting the order in, make sure to specify it needs to be taken with a substantial snack or meal (<u>500 calories</u>) ● Max IM dose of 40 mg/day
Mood Stabilizers	Divalproex Sodium and Valproate (Depakote ®, Depacon ®, Depakene ®, Depakote ER ®, Depakote Sprinkles ®)	Hepatic encephalopathy Benign hepatic transaminase elevations Asymptomatic leukopenia and thrombocytopenia Comes as IR capsule, DR sprinkle capsule, IR oral solution, IV solution, DR tablet, ER tablet ● For <u>bipolar disorder</u>, there is not a huge difference between the formulations for efficacy ● For <u>seizure disorders</u>, make sure the check home dosage form	Hepatotoxicity Increased risk of valproate-induced liver failure in patients with hereditary neurometabolic syndromes caused by DNA mutations Teratogenic - neural tube defects, decreased IQ scores Life-threatening pancreatitis	Labs/Diagnostics: ● Pregnancy test if starting ● Ammonia ● VPA level (compliance/toxicity) ○ Target VPA level 50-125 ● LFTs ● CBC with diff
	Carbamazepine (Tegretol ®)	Skin rashes (SJS) Mild leukopenia	Serious dermatologic reactions and	● Autoinducer ● DRUG INTERACTIONS - always do an interaction check on

	Epitol ®, Equetro ®)	Mild liver enzyme elevations Mild thrombocytopenia Hyponatremia Agranulocytosis Aplastic anemia Thrombocytopenia Hepatic failure Exfoliative dermatitis (SJS) Pancreatitis Comes as ER capsule, oral suspension, tablet, chewable tablet, ER tablet	HLA-B*1502 allele Aplastic anemia and agranulocytosis	UpToDate or Micromedex Can draw level - Carbamazepine goal - 4-12 - Levels are important in seizure disorders (this is what was studied), not for mood stabilization in BPD Labs/Diagnostics: - HLA-B*1502 status - CBC with diff
	Lamotrigine (Lamictal ®)	Needs SLOW titration to avoid SJS No renal adjustments Hepatic - Moderate to severe without ascites: 25% decrease - Moderate to severe with ascites: 50% decrease Comes as tablets, ODT, chewable tablets, ER tablets (24 hours) Tablets come as kits for easier titration	SJS	- SLOW titration (25 mg daily for two weeks, 50 mg daily for next two weeks, etc.) - If the patient has missed more than <u>5 days</u> of the dose, the medication needs to be re-titrated from the starting dose - Monitor for rash that is accompanied by fever or sore throat or with prominent facial or muscle involvement - this is likely SJS and the medication should be discontinued Drug Interactions: - If using with <u>carbamazepine</u>, this increases the metabolism so the lamotrigine dose should be doubled - If using with <u>valproic acid</u>, this decreases the metabolism so lamotrigine should be halved - If using with <u>oral contraceptives</u>, this increases the metabolism so the lamotrigine dose should be increased
	Lithium (Lithobid ®)	Lithium levels (dose related toxicity) Comes as capsules, oral solution, tablets, ER tablets	Monitoring for lithium toxicity	Labs/Diagnostics: - Pregnancy test (if starting medication - contraindicated in first trimester (Ebstein's anomaly)) - Serum sodium - Lithium level (compliance/toxicity) - Target Level - 0.6-1.2 - Thyroid - BUN

	Oxcarbazepine	Does not induce its own metabolism like carbamazepine Comes as oral suspension, tablets and ER tablets	None	Labs: • Serum sodium • Thyroid function - pediatric patients • CBC
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TABLE 5. Antipsychotic receptor binding properties

	Trade name	D ₁	D ₂	D ₃	D ₄	D ₅	5-HT _{1A}	5-HT _{2A}	5-HT _{2C}	5-HT ₇	H ₁	Musc M ₁	α ₁	α ₂	Comments
First-generation antipsychotics															
	Chlorpromazine Thorazine	+	+++	+++	++	+	0	+++	++	++	+++	++	+++	+	
	Fluphenazine Prolixin	++	++++	++++	++	++	+	++	+	+++	++	0	+++	0	
	Haloperidol Haldol	+	+++	+++	+++	+	0	++	0	+	0	0	++	0	
	Loxapine Loxitane	++	++	++	+++	++	0	+++	++	++	+++	+	++	0	
	Molindone Moban	0	++	++	0		0	0	0	0	0	0	0	+	
	Perphenazine Trilafon	++	++++	++++	++		0	+++	+	++	+++	0	++	+	
	Pimozide Orap	0	++++	+++	++		+	++	0	++++	+	+	+	+	Moderate activity at dopamine transporter
	Thioridazine Mellaril	++	++	+++	++	+	+	++	++	++	++	+++	+++	+	
	Thiothixene Navane	+	++++	++++	+	+	+	++	0	++	+++	0	++	0	
	Trifluoperazine Stelazine	+	+++	++++	++		+	++	+	+	++	+	++	0	
Second-generation antipsychotics															
	Aripiprazole Abilify	+	////	+++	+	0	///	+++	++	++	++	0	++	+	
	Asenapine Saphris, Secuado	+++	+++	++++	+++		+++	++++	++++	++++	+++	0	+++	+++	
	Brexipiprazole Rexulti	+	///	+++	++++		////	++++	++	+++	++	0	+++	++++	
	Cariprazine Vraylar		///	++++			///	++	+	+	++	0	+		
	Clozapine Clozaril, FazaClo, Versacloz	+	+	+	++	+	/	+++	++	++	+++	///	+++	+	
	Iloperidone Fanapt	+	++	++	++	+	//	++++	++	++	+	0	+++	+++	
	Lurasidone Latuda	+	+++	++	++		/	++++	+	++++	0	0	++	++	
	Olanzapine Zyprexa	++	++	++	++	++	0	+++	++	+	+++	+++	++	+	
	Paliperidone Invega	+	+++	+++	++	++	+	++++	++	+++	+++	0	+++	++	
	Quetiapine Seroquel	0	+	+	0	0	/	+	0	+	+++	+	++	0	
	Risperidone Risperdal	+	+++	+++	+++	+	+	++++	++	+++	++	0	+++	+++	
	Ziprasidone Geodon	+	+++	+++	++	+	///	++++	++++	+++	++	0	+++	+	Weak activity at norepinephrine and serotonin transporter

Note. ++++=very strong binding ($K_i < 1$ nM); +++=strong binding ($1 \text{ nM} \leq K_i < 10$ nM); ++=moderate binding ($10 \text{ nM} \leq K_i < 100$ nM); +=weak binding ($100 \text{ nM} \leq K_i < 1,000$ nM); 0=very weak or negligible binding ($K_i \geq 1,000$ nM). For partial agonists, / is used instead of + to denote relative binding values.

TABLE 6. Antipsychotic medications: relative side effects of oral formulations

	Trade name	Akathisia	Parkinsonism	Dystonia	Tardive dyskinesia	Hyperprolactinemia ^a	Anticholinergic	Sedation
First-generation antipsychotics								
Chlorpromazine	Thorazine	++	++	++	+++	+	+++	+++
Fluphenazine	Prolixin	+++	+++	+++	+++	+++	+	+
Haloperidol	Haldol	+++	+++	+++	+++	+++	+	+
Loxapine	Loxitane	++	++	++	++	++	++	++
Molindone	Moban	++	++	++	++	++	+	++
Perphenazine	Trilafon	++	++	++	++	++	++	++
Pimozide	Orap	+++	+++	++	+++	+++	+	+
Thioridazine	Mellaril	+	+	+	+	++	+++	+++
Thiothixene	Navane	+++	+++	+++	+++	+++	+	+
Trifluoperazine	Stelazine	++	++	++	++	++	++	+
Second-generation antipsychotics								
Aripiprazole	Abilify	++	+	+	+	+	+	+
Asenapine	Saphris	++	+	++	++	++	+	++
Brexipiprazole	Rexulti	++	+	+	+	+	+	++
Cariprazine	Vraylar	++	+	+	+	+	++	++
Clozapine	Clozaril, FazaClo, Versacloz	+	+	+	+	+	+++	+++
Iloperidone	Fanapt	+	+	+	+	++	+	++
Lurasidone	Latuda	++	++	++	++	+	+	++
Olanzapine	Zyprexa	++	++	+	+	++	++	+++
Paliperidone	Invega	++	++	++	++	+++	+	+
Quetiapine	Seroquel	+	+	+	+	+	++	+++
Risperidone	Risperdal	++	++	++	++	+++	+	++
Ziprasidone	Geodon	++	+	+	+	++	+	++

TABLE 6. Antipsychotic medications: relative side effects of oral formulations (continued)

	Trade name	Seizures	Orthostasis	QT prolongation	Weight gain	Hyperlipidemia	Glucose abnormalities	Comments
First-generation antipsychotics								
Chlorpromazine	Thorazine	++	+++	+++	++	+	++	
Fluphenazine	Prolixin	+	+	++	++	+	+	
Haloperidol	Haldol	+	+	++	++	+	+	
Loxapine	Loxitane	+	++	++	+	+	+	
Molindone	Moban	+	+	++	+	+	+	
Perphenazine	Trilafon	+	++	++	++	+	+	
Pimozide	Orap	+++	+	+++	+	+	+	
Thioridazine	Mellaril	++	+++	+++	++	+	+	Pigmentary retinopathy; high rates of sexual dysfunction; avoid use if QTc interval is >450 msec or with concomitant use of drugs that prolong the QTc interval or inhibit CYP2D6
Thiothixene	Navane	+++	+	++	+	+	+	
Trifluoperazine	Stelazine	+	+	++	++	+	+	
Second-generation antipsychotics								
Aripiprazole	Abilify	+	+	+	+	+	+	FDA safety alert for impulse control disorders (e.g., gambling, binge eating); may reduce hyperprolactinemia with other antipsychotics
Asenapine	Saphris	+	++	++	++	++	++	Oral hypoesthesia
Brexpiprazole	Rexulti	+	+	++	+	++	+	
Cariprazine	Vraylar	+	+	++	++	+	+	

TABLE 6. Antipsychotic medications: relative side effects of oral formulations (continued)

	Trade name	Seizures	Orthostasis	QT prolongation	Weight gain	Hyperlipidemia	Glucose abnormalities	Comments
Second-generation antipsychotics (continued)								
Clozapine	Clozaril, FazaClo, Versacloz	+++	+++	++	+++	+++	+++	Increased salivation common; high rate of sexual dysfunction; severe constipation and paralytic ileus possible; fever can occur with initiation; myocarditis is infrequent; cardiomyopathy and severe neutropenia are rare
Iloperidone	Fanapt	+	+++	+++	++	+	++	Dose-related creatinine increase in some patients
Lurasidone	Latuda	+	+	+	+	++	++	
Olanzapine	Zyprexa	++	++	++	+++	+++	+++	
Paliperidone	Invega	+	++	++	++	++	+	Intraoperative floppy iris syndrome reported
Quetiapine	Seroquel	++	++	++	++	+++	++	
Risperidone	Risperdal	+	++	++	++	+	++	
Ziprasidone	Geodon	+	++	+++	+	+	+	

^aIn general, rates of sexual dysfunction parallel rates of hyperprolactinemia except where noted in comments.

Note. +=seldom; ++=sometimes; +++=often. CYP=cytochrome P450.

Source. Credible Meds 2019; Hirsch et al. 2017; La Torre et al. 2013; Lexicomp 2019; Micromedex 2019; Pisani et al. 2002; Procyshyn et al. 2019; van Dijk et al. 2018.